

BPL-700 Magnet Circuit

Spacer height sweep, coil geometry, and power-supply requirements

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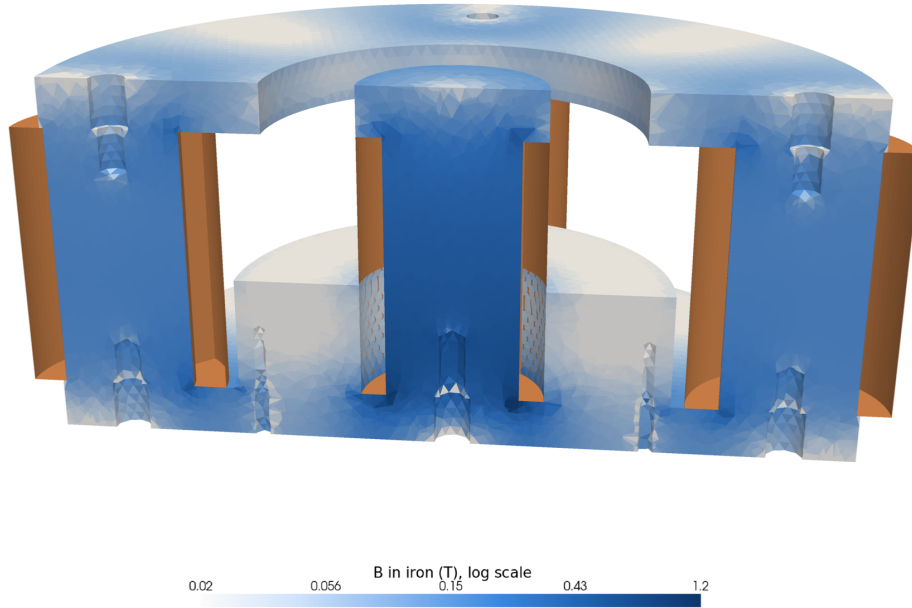


Figure 1: BPL-700 magnet circuit, half-section through the thrust axis and one outer core, baseline geometry (spacer 0.875 in), at the 300 G operating point. Iron is coloured by $|B|$; copper is the five windings. The exit (top plate) is at the top. Nonlinear solve, A36-proxy iron.

Summary

This report sizes the `chamber_spacer`, the A36 ring between the chamber's anode end and the bottom plate. Its height is set by `emag_height` (`spacer = emag_height - 1.125` in), which is also the height of every coil and core, so a shorter spacer means shorter windings. The inner winding cannot grow outward (the chamber bore sits on it), only inward by thinning the core. 25 full-3D magnetostatic solves cover spacer heights 0–0.875 in in 0.125 in steps at three inner-core diameters (1.000, 0.875, 0.750 in), each scaled to exactly 300 G radial field at the channel exit. It reports the resulting coil geometries, windings and supply requirements; choosing among them is left to the reader.

- **The spacer height hardly changes the field per ampere-turn.** Across the whole sweep the inner coil needs 326–339 A-turns (outer coils 207–215 each) for 300 G, a 4% spread, of which about $\pm 1\%$ is mesh noise. The circuit is gap-dominated, and the exit gap does not move with the spacer.
- **It changes the winding a great deal.** Shorter coils hold fewer turns, so the same A-turns need more current. At the current 1.000 in core, AWG 20 goes from 1.27 A / 5.1 W / 61°C at the full 0.875 in spacer

to 2.25 A / 10.0 W / 95°C with no spacer.

- **Thinning the core buys that room back almost for free.** A 0.750 in core costs about 2% more A-turns but nearly doubles the inner winding window. *No spacer with a 0.750 in core needs 1.28 A and 5.4 W, essentially today's 1.27 A and 5.1 W.*
- **Nothing approaches saturation.** The highest core flux density anywhere in the sweep is 0.74 T (0.750 in core, 0.875 in spacer), against a knee of 1.3–1.5 T on the measured 1020 curve. Core flux *rises* with spacer height: the steel spacer wraps the inner coil and pulls leakage flux through the core.
- **Shorter spacers shape the field slightly better.** The field at the anode end drops from 43 G to 22 G (1.000 in core) as the spacer goes from 0.875 in to 0, closer to the target lens profile, at a cost of about one point of radial purity (97.0% to 95.9%).
- **Supply.** One constant-current channel for the whole series string. With AWG 20 every swept design runs at 0.72–2.25 A and under 4.5 V hot (under 10 W), so a supply rated ≥ 2.8 A and ≥ 9 V covers them all; other gauges trade current for voltage at the same power (Sec. 5).

1 Configuration and constraints

The circuit is one inner pole and four outer poles between two 0.375 in iron plates (Fig. 2). The top plate is the exit pole piece; the inner core carries a 1.375 in pole cap flush with it, and the radial gap between that cap and the top-plate bore is what puts the field across the discharge channel. Flux returns through the four outer cores and the bottom plate. Inner and outer windings are driven with opposite polarity from one series current.

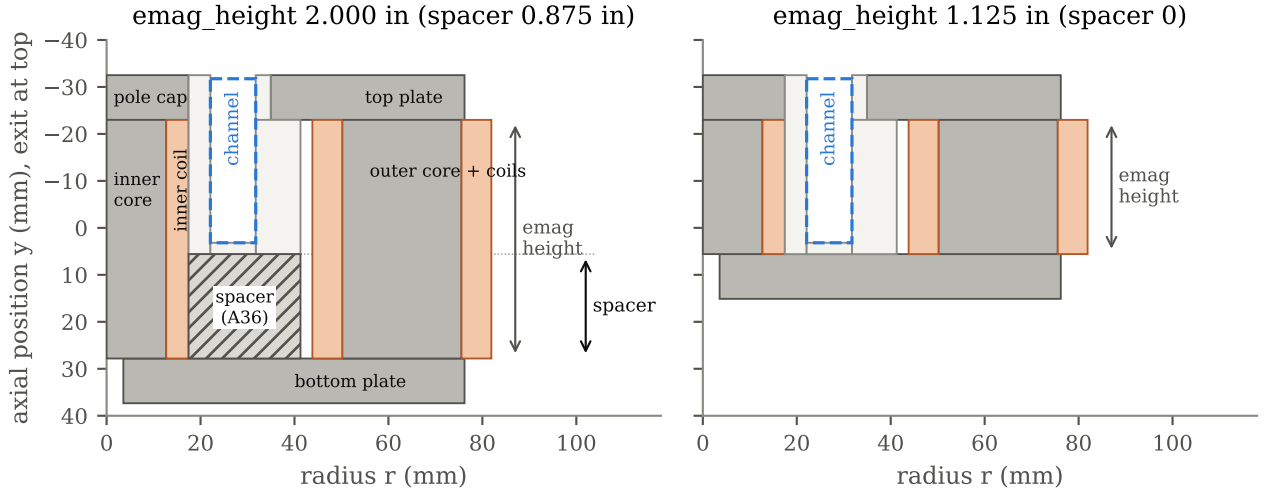


Figure 2: Half-section in the plane of one outer core, at the two ends of the sweep. Grey: iron (A36 plates and spacer; A108 cores). Hatched: `chamber_spacer`. Orange: windings. Light: chamber (PTFE). Dashed blue: the plasma channel used for every field statistic in this report (r 22.10–31.75 mm, anode at $y = +3.18$ mm, exit at $y = -31.75$ mm).

Measured directly from the Onshape STEP exports, not assumed:

- **The spacer is fixed by `emag_height`.** The chamber and top plate never move; the chamber's anode end sits at $y = +5.59$ mm and `emag_height` moves only the bottom plate. So

$$h_{\text{spacer}} = \text{emag_height} - 1.125 \text{ in},$$

and $h_{\text{spacer}} \geq 0$ means `emag_height` ≥ 1.125 in. At exactly 1.125 in Onshape drops the spacer body and the bottom plate sits on the chamber. All coils and cores are `emag_height` tall.

- **The inner coil cannot grow outward.** The chamber bore and the spacer bore are both $r = 17.46$ mm, exactly the inner coil's outer radius (OD 1.375 in). With the chamber fixed, the inner winding window is $4.76 \text{ mm} \times \text{emag_height}$ at the current 1.000 in core. It can only be recovered by thinning the core under the winding (`inner_coil_id`); the pole cap stays at 1.375 in regardless.
- **The outer coils have room.** Outer core axes sit 62.86 mm off-axis and the chamber's outer wall is at $r = 41.28$ mm, so an outer coil can reach ≈ 1.70 in OD before touching it (currently 1.5 in).

- **Stock thicknesses.** The spacer is a single laser-cut A36 part only at a stocked thickness: 0.187, 0.250, 0.313, 0.375 or 0.500 in. Those are marked [†] in the tables.

2 Materials

Part	Material	Source	Magnetic model
Top, bottom plates; spacer	ASTM A36, hot-rolled P&O	SendCutSend	AISI 1008 curve as proxy (secondhand)
Inner, outer emag cores	ASTM A108 cold-finished bar (taken as 1018)	McMaster	measured 1020 curve, Fermilab TM-1197
Windings	fiberglass-served Cu magnet wire	–	$\mu_r = 1$; current density only
Chamber	PTFE	–	$\mu_r = 1$ (not meshed)

Table 1: Materials. Full properties, sources and confidence ratings are in `MATERIALS.md`.

The circuit is gap-dominated, so the channel field barely depends on the iron grade (Sec. 3). Grade does set where a thinned core saturates, which is why the cores are judged against a measured curve (Fig. 7, right). Cold drawing lowers permeability slightly relative to the measured stock; a stress-relief anneal after machining recovers it if a core ever runs near the knee.

3 Method

Geometry. Each point is a full-assembly STEP export from the parametrised Onshape document (5 configuration parameters), imported, meshed and solved in 3D by the Gmsh/GetDP pipeline (magnetic vector potential, edge elements, $\sim 5.9 \times 10^5$ unknowns). The chamber is fixed, so the channel region is taken from its CAD-confirmed dimensions rather than re-detected per run.

Linear iron, and why that is enough. Every sweep point uses linear iron ($\mu_r = 1500$), about 2.5 min per solve against ~ 70 min nonlinear. Checked, not assumed: on the baseline geometry the linear solve reads +1.2% on the exit field relative to the committed nonlinear (A36-proxy) solve at equal current. A second check re-solved the highest-core-flux geometry (0.750 in core, 0.875 in spacer) with nonlinear iron. Its half-current stage converged (9 Picard iterations), but the full-current stage stalled at a relative residual of 0.11 instead of reaching 10^{-4} , so *it is not a validated solution*. Taken for what it is, its field agrees with the linear solve to 0.3% on the exit field and 0.5% in the outer cores, and puts the inner-core shaft flux density 3.4% *lower* (the linear figures are conservative there). The one sizeable disagreement is the 99.9th-percentile iron field (13% higher), i.e. a small number of high-field corner elements, which is plausibly where the Picard iteration is stuck. A converged check needs a finer load ramp or a Newton iteration. Because the model is linear, each geometry needs one solve: the field scales with ampere-turns, so the A-turns for exactly 300 G at the exit follow by scaling, and so do the iron flux densities.

Excitation. Inner and outer A-turns are held at the committed ratio (338 : 215 per coil, the field-direction optimum from the earlier current-ratio sweep) and scaled together to put the channel-averaged B_r at the exit station at 300 G. Sweeping the ratio per geometry is a cheap follow-up (two basis solves each) if the shape results below suggest it.

Metrics. The channel is split into 20 axial stations from anode to exit; B_r and B_{axial} are volume-element averages over each station. Radial purity is the mean of $B_r/|B|$. Core flux density is the mean axial B across the shaft cross-section (inside 90% of the core radius), taken at its maximum along the shaft, since leakage flux joins the core along its length.

Windings. `sizing/winding_design.py` turns A-turns into a winding. Hexagonally nested round wire; insulated diameters are single-glass-served estimates. The inner coil is wound full; the outer coils are wound to whatever turn count gives the required A-turn ratio at the same series current (or full, if the outer window runs out first). Resistance uses the radius the copper actually occupies. It reproduces the committed 260/165-turn AWG 20 design exactly.

Thermal. `thermal/radiation_balance.py`: vacuum, radiation from the outer envelope only (taken from each geometry’s own mesh), emissivity 0.30, coils bonded to their cores. Temperatures feed back into copper resistance.

4 Results

4.1 Field in the circuit

Fig. 3 shows the flux path at both ends of the sweep: up the inner core, radially across the channel from the pole cap to the top-plate bore, down the outer cores and back through the bottom plate. The field in the channel is set by that exit gap, which is identical in every configuration, so the two maps differ mainly behind the channel. With a long spacer, the steel spacer sits just 4.8 mm from the inner core, across the inner winding, and offers flux a path back to the bottom plate that bypasses the channel (this is what loads the core; Sec. 4.4). With no spacer, the bottom plate sits 2.4 mm behind the anode face instead.

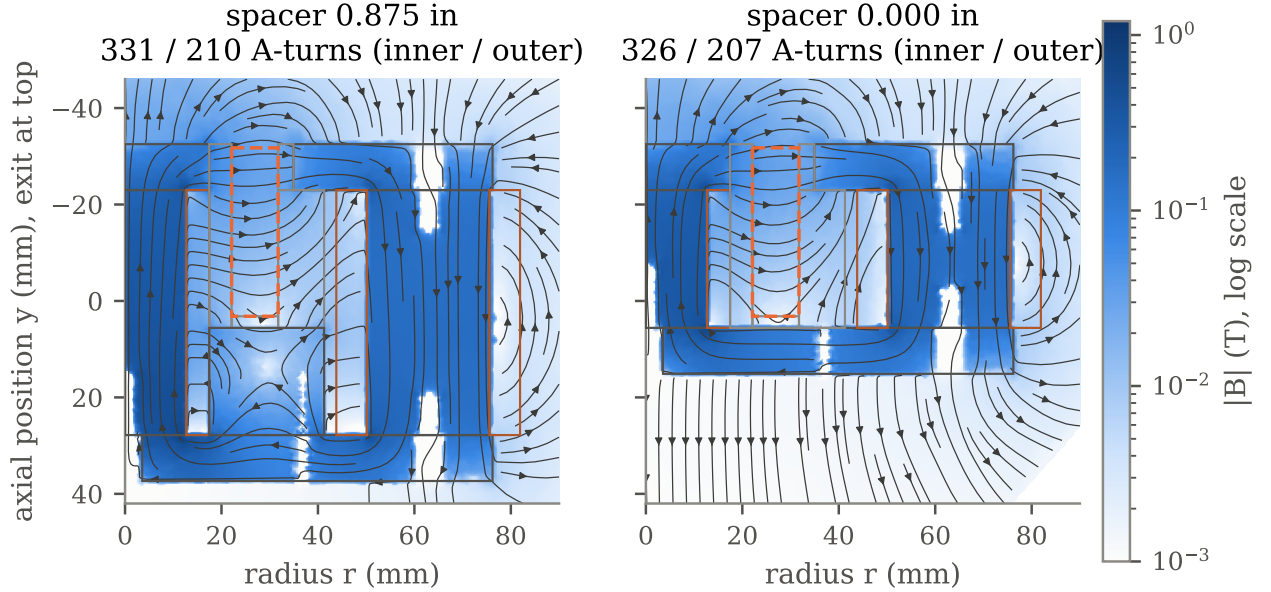


Figure 3: $|B|$ (log scale) and field lines in the plane through the axis and two opposite outer cores, at each geometry’s own 300 G excitation, 1.000 in core. Left: longest spacer swept (0.875 in). Right: no spacer. Outlines: iron, windings, chamber; dashed orange: plasma channel.

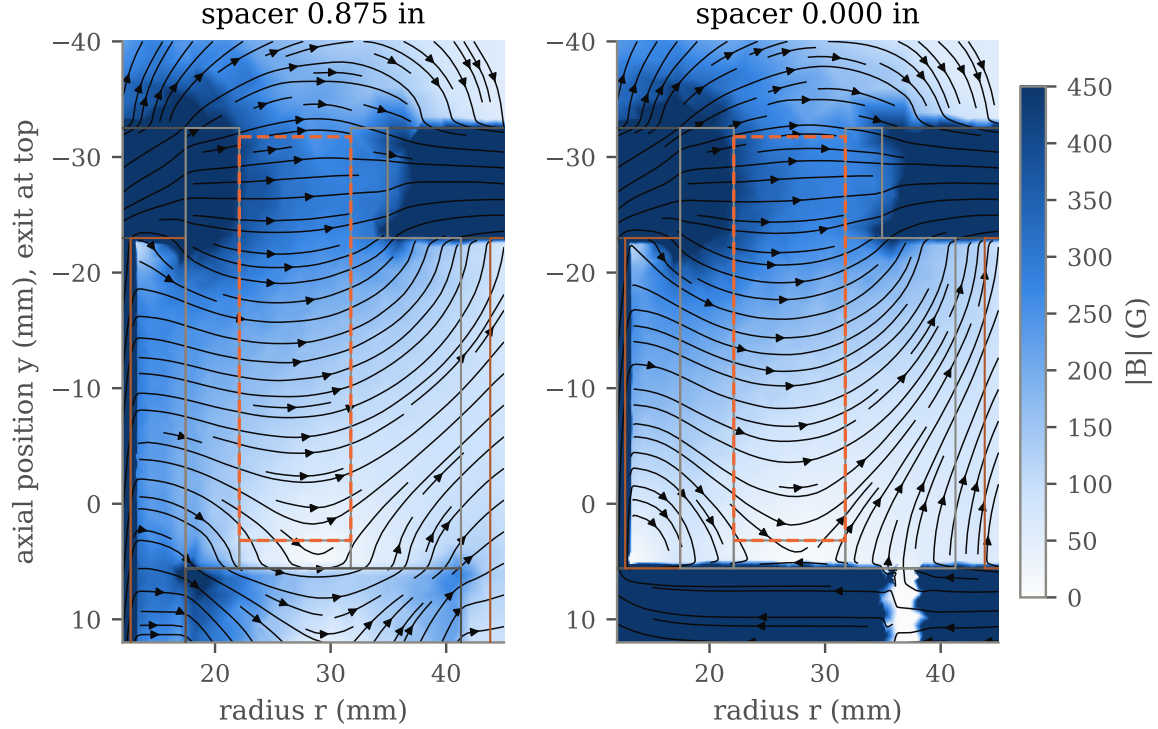


Figure 4: Close-up of the channel (dashed), linear colour scale in gauss, same two geometries. The field crosses the channel radially from the inner pole cap to the top-plate bore and weakens toward the anode.

4.2 Channel field shape

Every configuration keeps the same basic lens shape: B_r rises from the anode to a peak of about 317 G roughly 4 mm inside the exit plane, and is 300 G at the exit by construction (Fig. 5). The spacer only moves the anode half of the profile. Shorter spacers pull the anode-end field down (43 G at 0.875 in to 22 G at 0, 1.000 in core), which moves it toward the target lens and away from the anode, and they reduce the axial component there. The trade is a slightly less radial field overall: radial purity falls from 97.0% to 95.9% (Table 3). A thinner core shifts the anode field by only a few gauss. All configurations still overshoot the idealised target in the middle of the channel; that is a property of this pole geometry, not of the spacer.

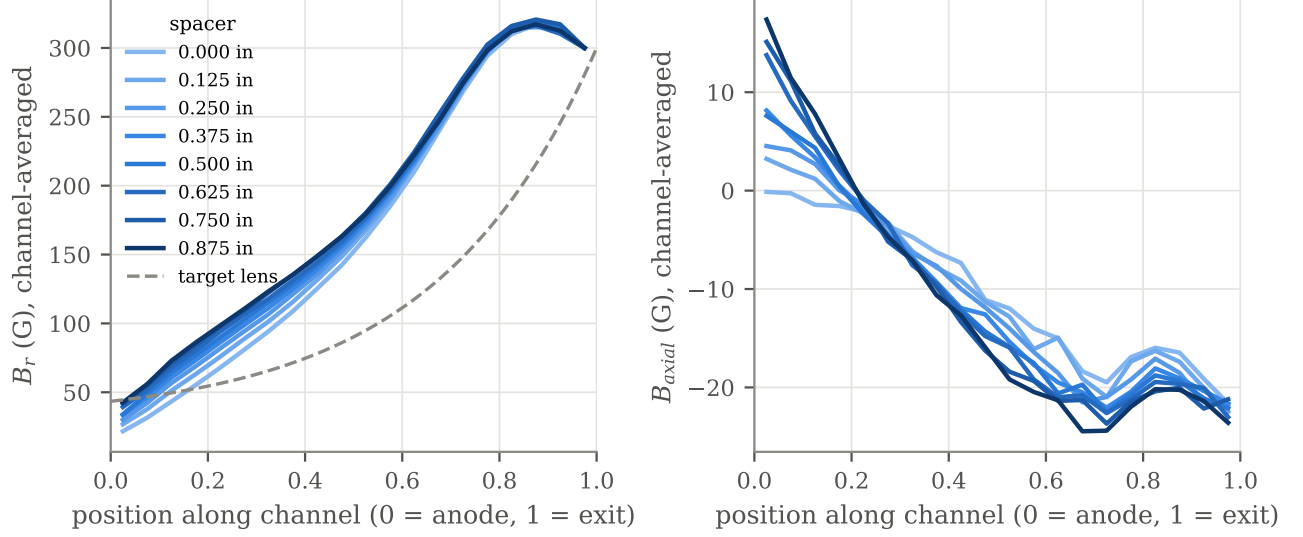


Figure 5: Channel-averaged radial (left) and axial (right) field from anode to exit, every spacer height, 1.000 in core, each at its own 300 G excitation. Dashed grey: the target magnetic-lens profile used by `field_quality.py`.

4.3 Ampere-turns needed

The A-turns needed for 300 G barely move (Fig. 6). Removing the spacer entirely changes the requirement by -1.3% at the 1.000 in core, comparable to the $\pm 1\%$ scatter between neighbouring points, which is mesh noise (each geometry is re-meshed from scratch). The one systematic trend is the core: a 0.750 in core needs about 2% more A-turns than a 1.000 in one, as more of the inner coil's flux leaks past the thinner core. Growing the outer coils to 1.625 in OD (one check point, 1.5 in `emag_height`) changes the requirement by -0.4%, i.e. not at all: outer-coil OD is purely a winding-room choice.

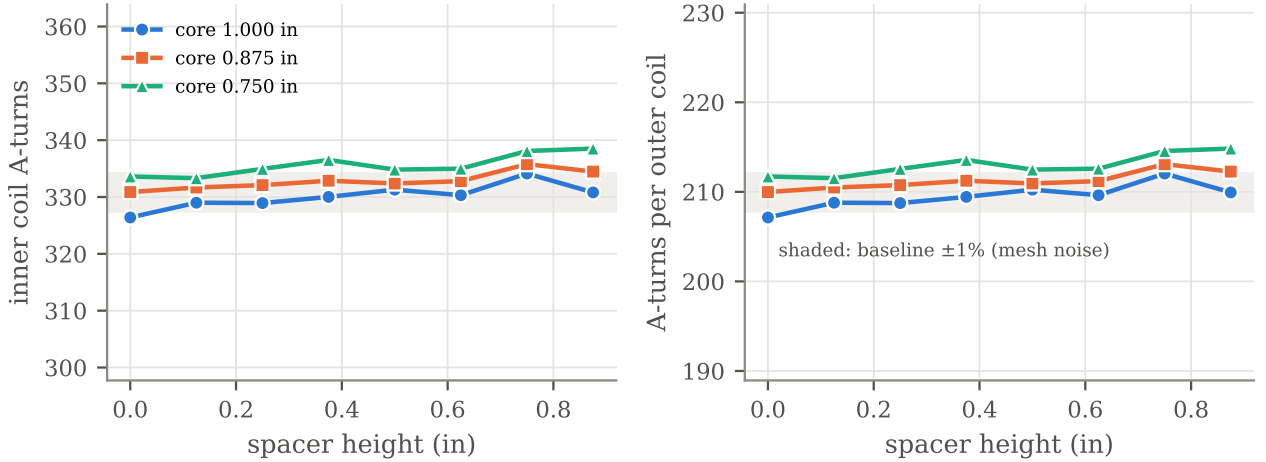


Figure 6: A-turns for 300 G at the channel exit, inner coil and each outer coil, and the resulting field per inner ampere-turn.

4.4 Core flux density

Every core stays far below the knee (Fig. 7). The inner core's shaft flux density runs 0.34–0.46 T at 1.000 in, 0.45–0.56 T at 0.875 in and 0.61–0.74 T at 0.750 in, scaling roughly with $1/d^2$ as it must for a fixed flux. The outer cores sit at 0.17–0.19 T throughout. Counter-intuitively, core flux is highest at the *longest* spacer. The

steel spacer wraps the inner coil and gives flux a return path through the bottom plate, so the lower part of the core carries that leakage on top of the useful channel flux. Shortening the spacer removes it. The worst case in the sweep, 0.74 T, is about half the 1.3–1.5 T knee; on the measured 1020 curve the cores stay on the steep, near-linear part of the curve everywhere swept.

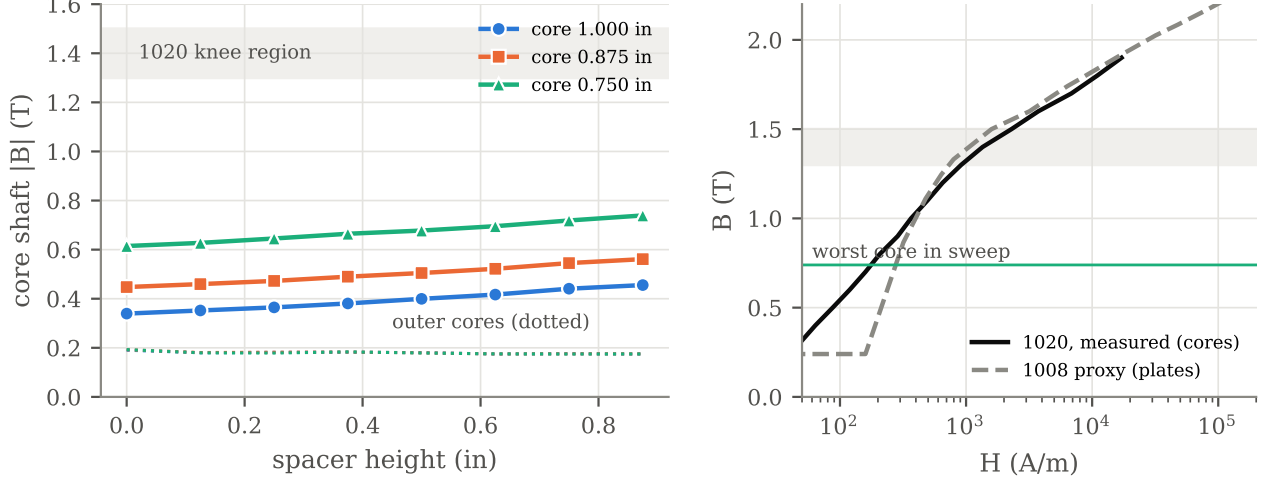


Figure 7: Left: highest shaft-averaged flux density along each core at 300 G, per inner-core diameter (solid: inner core; dotted: outer cores). Right: the measured 1020 curve used for the cores and the 1008 proxy used for the plates, with the worst swept core marked. Shaded: the knee.

4.5 Coil geometries and windings

Table 2 lists every coil geometry. Tables 3 and 4 give the magnetics and the AWG 20 winding for each; Appendix A has every gauge.

Three things stand out in Fig. 8:

- **Current and power track winding room, not the field.** The A-turns are almost constant, so current is inversely proportional to how many turns fit. The inner window is $(\text{core-to-chamber build}) \times \text{emag_height}$, and both a short spacer and a thick core shrink it.
- **String voltage is nearly constant** at 4.0–4.5 V hot across every AWG 20 design. For a fixed gauge $V = IR = (NI) \rho L_{\text{turn}} / A_{\text{wire}}$: it depends only on the A-turns and the mean turn length, not on how many turns fit. Since the A-turns hardly change, neither does the voltage, and power (VI) follows current.
- **A thinner core offsets a shorter spacer.** Each step from 1.000 to 0.875 to 0.750 in roughly offsets 0.4 in of spacer. No spacer with a 0.750 in core (261/166 turns, 1.28 A, 5.4 W, 69°C) matches today’s design (260/165 turns, 1.27 A, 5.1 W, 61°C) almost exactly.

Every configuration is thermally comfortable: the hottest coil is 95°C (1.000 in core, no spacer), against a 200°C wire-insulation limit.

spacer (in)	emag_height (in)	inner coil				outer coil (each)		
		ID (in)	OD (in)	build (mm)	window (mm ²)	ID (in)	OD (in)	window (mm ²)
0.000	1.125	1.000	1.375	4.76	136	1.000	1.500	181
0.125	1.250	1.000	1.375	4.76	151	1.000	1.500	202
0.250 [†]	1.375	1.000	1.375	4.76	166	1.000	1.500	222
0.375 [†]	1.500	1.000	1.375	4.76	181	1.000	1.500	242
0.500 [†]	1.625	1.000	1.375	4.76	197	1.000	1.500	262
0.625	1.750	1.000	1.375	4.76	212	1.000	1.500	282
0.750	1.875	1.000	1.375	4.76	227	1.000	1.500	302
0.875	2.000	1.000	1.375	4.76	242	1.000	1.500	323
0.000	1.125	0.875	1.375	6.35	181	1.000	1.500	181
0.125	1.250	0.875	1.375	6.35	202	1.000	1.500	202
0.250 [†]	1.375	0.875	1.375	6.35	222	1.000	1.500	222
0.375 [†]	1.500	0.875	1.375	6.35	242	1.000	1.500	242
0.500 [†]	1.625	0.875	1.375	6.35	262	1.000	1.500	262
0.625	1.750	0.875	1.375	6.35	282	1.000	1.500	282
0.750	1.875	0.875	1.375	6.35	302	1.000	1.500	302
0.875	2.000	0.875	1.375	6.35	323	1.000	1.500	323
0.000	1.125	0.750	1.375	7.94	227	1.000	1.500	181
0.125	1.250	0.750	1.375	7.94	252	1.000	1.500	202
0.250 [†]	1.375	0.750	1.375	7.94	277	1.000	1.500	222
0.375 [†]	1.500	0.750	1.375	7.94	302	1.000	1.500	242
0.500 [†]	1.625	0.750	1.375	7.94	328	1.000	1.500	262
0.625	1.750	0.750	1.375	7.94	353	1.000	1.500	282
0.750	1.875	0.750	1.375	7.94	378	1.000	1.500	302
0.875	2.000	0.750	1.375	7.94	403	1.000	1.500	323

Table 2: Coil geometry of every swept configuration. Inner coil ID = core diameter; OD is fixed at 1.375 in by the chamber bore. [†] spacer at a stocked A36 thickness.

spacer (in)	core (in)	A-turns for 300 G		B_r peak (G)	B_r anode (G)	radial purity (%)	shaft $ B $ (T)		iron p99.9 (T)
		inner	outer ea.				inner	outer	
0.000	1.000	326	207	317	22	95.9	0.34	0.19	0.42
0.125	1.000	329	209	318	27	96.4	0.35	0.18	0.44
0.250 [†]	1.000	329	209	315	30	96.6	0.36	0.18	0.45
0.375 [†]	1.000	330	209	318	33	96.8	0.38	0.18	0.47
0.500 [†]	1.000	331	210	320	34	96.9	0.40	0.18	0.49
0.625	1.000	330	210	316	39	96.9	0.42	0.17	0.50
0.750	1.000	334	212	321	42	97.0	0.44	0.18	0.53
0.875	1.000	331	210	317	43	97.0	0.46	0.17	0.55
0.000	0.875	331	210	317	20	95.5	0.45	0.19	0.52
0.125	0.875	332	210	317	24	96.1	0.46	0.18	0.54
0.250 [†]	0.875	332	211	317	27	96.3	0.47	0.18	0.55
0.375 [†]	0.875	333	211	316	30	96.5	0.49	0.18	0.56
0.500 [†]	0.875	332	211	318	30	96.7	0.50	0.18	0.58
0.625	0.875	333	211	317	35	96.8	0.52	0.17	0.60
0.750	0.875	336	213	319	37	96.9	0.55	0.18	0.62
0.875	0.875	334	212	318	38	96.9	0.56	0.18	0.63
0.000	0.750	334	212	317	19	94.9	0.61	0.19	0.68
0.125	0.750	333	212	318	23	95.6	0.63	0.18	0.70
0.250 [†]	0.750	335	213	318	25	95.9	0.65	0.18	0.72
0.375 [†]	0.750	337	214	319	27	96.1	0.66	0.18	0.73
0.500 [†]	0.750	335	212	317	28	96.3	0.68	0.18	0.74
0.625	0.750	335	213	317	32	96.5	0.70	0.18	0.76
0.750	0.750	338	215	317	33	96.6	0.72	0.17	0.79
0.875	0.750	339	215	318	34	96.7	0.74	0.17	0.80

Table 3: Magnetics per configuration, at each one's 300 G excitation (linear iron).

spacer	core	inner		outer (each)		I	R_{20}	V_{hot}	P_{hot}	T_{coil}	wire
(in)	(in)	turns	layers	turns	layers	(A)	(Ω)	(V)	(W)	($^{\circ}\text{C}$)	(m)
0.000	1.000	145	5	93	4	2.25	1.53	4.46	10.0	95	47
0.125	1.000	165	5	105	4	1.99	1.73	4.36	8.7	87	53
0.250 [†]	1.000	180	5	115	4	1.83	1.90	4.29	7.8	81	58
0.375 [†]	1.000	195	5	124	4	1.69	2.05	4.22	7.1	76	63
0.500 [†]	1.000	210	5	134	4	1.58	2.21	4.19	6.6	72	68
0.625	1.000	230	5	146	4	1.44	2.41	4.10	5.9	67	74
0.750	1.000	245	5	156	4	1.36	2.58	4.12	5.6	64	79
0.875	1.000	260	5	165	4	1.27	2.73	4.03	5.1	61	84
0.000	0.875	203	7	129	5	1.63	2.14	4.29	7.0	78	66
0.125	0.875	231	7	147	5	1.44	2.44	4.21	6.0	71	75
0.250 [†]	0.875	252	7	160	5	1.32	2.66	4.15	5.5	67	82
0.375 [†]	0.875	273	7	174	5	1.22	2.89	4.12	5.0	63	89
0.500 [†]	0.875	294	7	187	5	1.13	3.11	4.06	4.6	60	96
0.625	0.875	322	7	205	5	1.03	3.41	4.02	4.2	56	105
0.750	0.875	343	7	218	5	0.98	3.62	4.03	3.9	55	112
0.875	0.875	364	7	231	5	0.92	3.84	3.97	3.7	52	118
0.000	0.750	261	9	166	6	1.28	2.77	4.22	5.4	69	86
0.125	0.750	297	9	189	6	1.12	3.16	4.13	4.6	63	97
0.250 [†]	0.750	324	9	206	6	1.03	3.44	4.10	4.2	59	106
0.375 [†]	0.750	351	9	223	6	0.96	3.73	4.08	3.9	56	115
0.500 [†]	0.750	378	9	240	6	0.89	4.01	4.02	3.6	53	124
0.625	0.750	414	9	263	6	0.81	4.40	3.98	3.2	51	136
0.750	0.750	441	9	280	6	0.77	4.68	4.00	3.1	49	144
0.875	0.750	468	9	297	6	0.72	4.97	3.98	2.9	47	153

Table 4: AWG 20 winding for every configuration (the committed gauge), series string of all five coils. Other gauges: Appendix A.

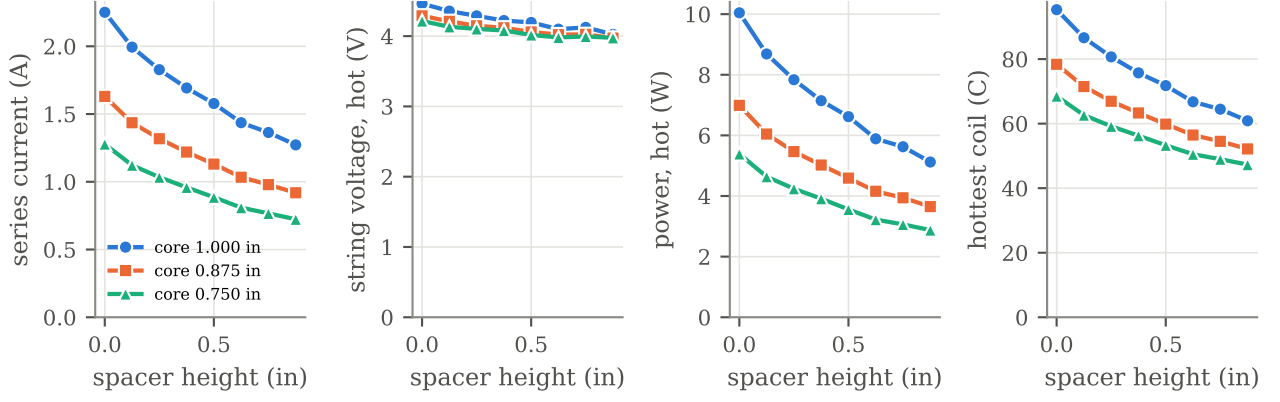


Figure 8: AWG 20: series current, string voltage and power at operating temperature, and hottest-coil temperature, per spacer height and core diameter.

5 Power-supply requirements

Nothing in the magnetics depends on the gauge: only the ampere-turn product reaches the field. The gauge sets how that product splits into current and voltage, and therefore which supply fits. Table 5 gives, for each gauge, the envelope over *every* swept geometry, so a supply chosen from it covers whichever spacer height is picked. Fig. 9 shows the same designs as operating points.

AWG	current (A)	max voltage, hot (V)	max power (W)	string L (mH)	suggested supply rating
16	1.61–5.73	1.7	10.0	4–51	≥ 7.2 A, ≥ 3 V
18	1.12–3.40	2.7	9.3	11–104	≥ 4.2 A, ≥ 5 V
20	0.72–2.25	4.5	10.0	26–252	≥ 2.8 A, ≥ 9 V
22	0.48–1.51	7.0	10.6	58–570	≥ 1.9 A, ≥ 14 V
24	0.33–0.95	11.4	10.8	146–1183	≥ 1.2 A, ≥ 23 V

Table 5: Supply envelope per wire gauge across all swept geometries (one series string, all five coils). “Hot” is at the steady-state coil temperature. Suggested rating: $1.25\times$ the highest current (room for the linear-model and wire-diameter uncertainty and a field margin above 300 G) and $2\times$ the highest hot voltage (room for lead resistance and ramping into the inductance).

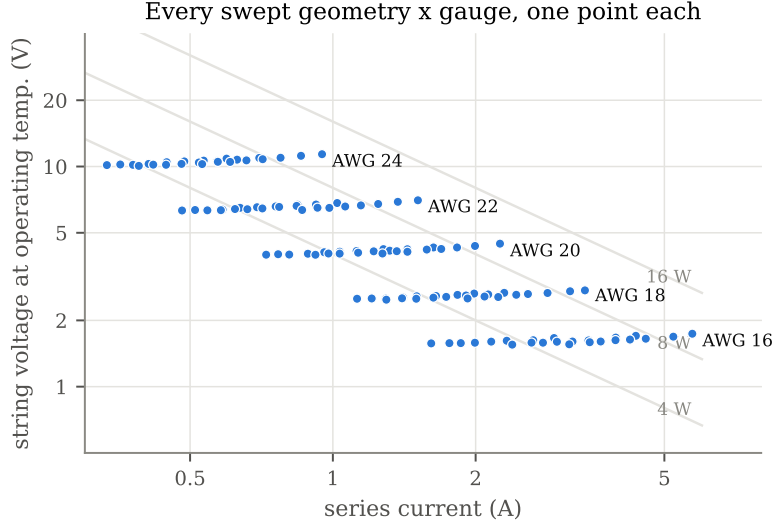


Figure 9: Every swept geometry in every gauge as a (current, voltage) operating point, with lines of constant power. Gauge moves a design along a constant-power line; geometry moves it across the lines.

What the supply needs, independent of gauge:

- **Constant-current (CC) regulation.** The field follows current, not voltage; copper resistance rises about 0.4%/K as the coils warm, so a constant-voltage supply would let the field drift down by the same amount. Any bench supply with a CC mode works; the load is a plain resistor-inductor.
- **One channel is enough.** All five coils run in series from one current. The inner/outer ratio is set by turn counts, not by separate supplies. A second supply (or a trim resistor on one leg) is only needed to tune the ratio in the lab.
- **Inductance, and switching off.** Stored field energy at the operating point is 66 mJ (from the nonlinear baseline solve), independent of gauge, so the string inductance is $L = 2W/I^2$ (Table 5). Ramp the current up and down rather than stepping it, and put a flyback diode (or TVS) across the string so an open circuit or relay under current cannot generate a large voltage spike. The time constant L/R is tens of milliseconds, so ripple is heavily filtered and does not matter.
- **Polarity.** The inner winding is driven opposite to the outer windings (flux crosses the channel from the inner pole to the top plate). Wire the string so the inner coil's sense is reversed relative to the four outer coils, and check it with a gaussmeter at first power-up: wrong polarity collapses the channel field.
- **Floating output.** The coils sit next to the discharge; an isolated (floating) output keeps the magnet circuit independent of the discharge supply's reference.

6 Limitations and open items

- **Insulated wire diameters are estimates.** They set turn counts, which can move by 10–20% with a real datasheet. That moves current and voltage (and the supply choice), not the field and barely the power.
- **Linear iron.** Each point is a linear solve. It is checked against a converged nonlinear solve only at the baseline geometry; the worst-case nonlinear check did not converge (Sec. 3), though its unconverged field agrees with the linear one in the channel and cores. The plates' B-H curve is a proxy (1008 for A36). The circuit is gap-dominated, so this mostly matters for the core saturation margin, which is judged against the measured 1020 curve.
- **Fixed inner/outer A-turn ratio.** Re-optimising the ratio per geometry could recover some field shape at short spacers.
- **The 300 G target is not derived** from the plasma requirements (the top-level sizing script assumes 150 G). The results scale linearly: halving the target halves every A-turn and current figure and quarters every power.
- **Magnets-only thermal.** No plasma heat load; emissivity 0.30 assumed (`thermal/README.md` has the

sensitivity).

A Every gauge, every geometry

spacer (in)	core (in)	AWG	turns in/out	I (A)	R_{hot} (Ω)	V_{20} (V)	V_{hot} (V)	P_{hot} (W)	T_{coil} ($^{\circ}\text{C}$)	J (A/mm ²)
0.000	1.000	16	57/37	5.73	0.30	1.35	1.74	10.0	95	4.4
		18	96/61	3.40	0.81	2.14	2.74	9.3	91	4.1
		20	145/93	2.25	1.98	3.45	4.46	10.0	95	4.3
		22	216/138	1.51	4.65	5.38	7.02	10.6	98	4.6
		24	344/219	0.95	11.99	8.69	11.37	10.8	99	4.6
0.125	1.000	16	63/40	5.22	0.32	1.34	1.69	8.8	87	4.0
		18	104/66	3.16	0.86	2.16	2.71	8.6	86	3.8
		20	165/105	1.99	2.18	3.46	4.36	8.7	87	3.9
		22	240/153	1.37	5.04	5.42	6.91	9.5	91	4.2
		24	384/244	0.86	13.06	8.75	11.19	9.6	91	4.2
0.250 [†]	1.000	16	72/46	4.57	0.36	1.34	1.65	7.5	79	3.5
		18	116/74	2.84	0.94	2.16	2.66	7.6	79	3.4
		20	180/115	1.83	2.35	3.47	4.29	7.8	81	3.5
		22	264/168	1.25	5.42	5.41	6.76	8.4	84	3.8
		24	424/270	0.78	14.13	8.76	10.96	8.5	84	3.8
0.375 [†]	1.000	16	78/50	4.23	0.39	1.35	1.64	6.9	75	3.2
		18	128/82	2.58	1.02	2.18	2.64	6.8	74	3.1
		20	195/124	1.69	2.49	3.47	4.22	7.1	76	3.3
		22	288/183	1.15	5.81	5.42	6.66	7.6	78	3.5
		24	464/295	0.71	15.17	8.78	10.79	7.7	79	3.5
0.500 [†]	1.000	16	84/54	3.94	0.41	1.36	1.63	6.4	71	3.0
		18	136/87	2.44	1.07	2.18	2.61	6.4	70	3.0
		20	210/134	1.58	2.66	3.49	4.19	6.6	72	3.0
		22	312/198	1.06	6.20	5.44	6.58	7.0	74	3.3
		24	504/320	0.66	16.22	8.81	10.66	7.0	74	3.2
0.625	1.000	16	90/58	3.67	0.44	1.36	1.60	5.9	67	2.8
		18	148/94	2.23	1.14	2.17	2.55	5.7	66	2.7
		20	230/146	1.44	2.85	3.47	4.10	5.9	67	2.8
		22	336/214	0.98	6.60	5.44	6.49	6.4	70	3.0
		24	544/346	0.61	17.28	8.79	10.49	6.4	70	3.0
0.750	1.000	16	96/61	3.48	0.46	1.36	1.59	5.5	64	2.7
		18	160/102	2.09	1.23	2.20	2.57	5.4	63	2.5
		20	245/156	1.36	3.02	3.51	4.12	5.6	64	2.6
		22	360/229	0.93	7.00	5.49	6.50	6.0	67	2.8
		24	584/371	0.57	18.36	8.89	10.51	6.0	67	2.8
0.875	1.000	16	105/67	3.15	0.49	1.35	1.56	4.9	60	2.4
		18	172/110	1.92	1.31	2.18	2.52	4.8	59	2.3
		20	260/165	1.27	3.16	3.47	4.03	5.1	61	2.5
		22	384/244	0.86	7.37	5.43	6.35	5.5	63	2.6
		24	624/396	0.53	19.35	8.79	10.26	5.4	63	2.6
0.000	0.875	16	76/49	4.35	0.39	1.38	1.70	7.4	81	3.3
		18	144/92	2.30	1.16	2.21	2.67	6.1	73	2.8
		20	203/129	1.63	2.63	3.49	4.29	7.0	78	3.1
		22	324/206	1.02	6.68	5.56	6.83	7.0	78	3.1
		24	473/301	0.70	15.64	8.81	10.94	7.7	82	3.4
0.125	0.875	16	84/54	3.95	0.42	1.38	1.67	6.6	75	3.0
		18	156/99	2.13	1.23	2.21	2.62	5.6	69	2.6
		20	231/147	1.44	2.93	3.51	4.21	6.0	71	2.8
		22	360/229	0.92	7.29	5.58	6.72	6.2	72	2.8
		24	528/336	0.63	17.10	8.83	10.74	6.7	76	3.1

spacer (in)	core (in)	AWG	turns in/out	I (A)	R_{hot} (Ω)	V_{20} (V)	V_{hot} (V)	P_{hot} (W)	T_{coil} ($^{\circ}\text{C}$)	J (A/mm 2)
0.250 †	0.875	16	96/61	3.46	0.47	1.37	1.62	5.6	68	2.6
		18	174/111	1.91	1.36	2.22	2.59	5.0	64	2.3
		20	252/160	1.32	3.15	3.51	4.15	5.5	67	2.5
		22	396/252	0.84	7.90	5.59	6.62	5.6	67	2.6
		24	583/370	0.57	18.52	8.82	10.55	6.0	70	2.8
0.375 †	0.875	16	104/66	3.20	0.50	1.37	1.60	5.1	64	2.4
		18	192/122	1.73	1.48	2.22	2.56	4.4	60	2.1
		20	273/174	1.22	3.38	3.52	4.12	5.0	63	2.4
		22	432/275	0.77	8.51	5.60	6.55	5.0	63	2.4
		24	638/405	0.52	19.98	8.84	10.43	5.4	66	2.5
0.500 †	0.875	16	112/72	2.97	0.54	1.38	1.60	4.7	61	2.3
		18	204/130	1.63	1.56	2.22	2.54	4.1	57	2.0
		20	294/187	1.13	3.59	3.51	4.06	4.6	60	2.2
		22	468/297	0.71	9.08	5.58	6.45	4.6	60	2.2
		24	693/440	0.48	21.43	8.83	10.28	4.9	62	2.3
0.625	0.875	16	120/77	2.77	0.57	1.38	1.58	4.4	58	2.1
		18	222/141	1.50	1.67	2.21	2.51	3.8	54	1.8
		20	322/205	1.03	3.89	3.52	4.02	4.2	56	2.0
		22	504/320	0.66	9.69	5.59	6.40	4.2	57	2.0
		24	748/475	0.44	22.89	8.84	10.18	4.5	59	2.2
0.750	0.875	16	128/82	2.62	0.60	1.39	1.58	4.2	56	2.0
		18	240/153	1.40	1.80	2.24	2.52	3.5	52	1.7
		20	343/218	0.98	4.11	3.55	4.03	3.9	55	1.9
		22	540/343	0.62	10.31	5.64	6.41	4.0	55	1.9
		24	803/510	0.42	24.38	8.92	10.20	4.3	57	2.0
0.875	0.875	16	140/89	2.39	0.65	1.38	1.55	3.7	53	1.8
		18	258/164	1.30	1.92	2.23	2.48	3.2	49	1.6
		20	364/231	0.92	4.32	3.53	3.97	3.7	52	1.8
		22	576/366	0.58	10.91	5.62	6.33	3.7	52	1.8
		24	858/545	0.39	25.82	8.89	10.07	3.9	54	1.9
0.000	0.750	16	114/73	2.93	0.57	1.42	1.66	4.9	65	2.2
		18	168/107	1.99	1.33	2.23	2.65	5.3	68	2.4
		20	261/166	1.28	3.30	3.55	4.22	5.4	69	2.5
		22	396/252	0.84	7.96	5.61	6.71	5.6	70	2.6
		24	559/355	0.60	18.20	8.93	10.86	6.5	75	2.9
0.125	0.750	16	126/80	2.65	0.61	1.40	1.63	4.3	60	2.0
		18	182/116	1.83	1.43	2.23	2.61	4.8	64	2.2
		20	297/189	1.12	3.68	3.54	4.13	4.6	63	2.2
		22	440/280	0.76	8.69	5.61	6.59	5.0	65	2.3
		24	624/396	0.53	19.91	8.92	10.64	5.7	69	2.6
0.250 †	0.750	16	144/92	2.33	0.70	1.42	1.62	3.8	56	1.8
		18	203/129	1.65	1.56	2.24	2.58	4.3	59	2.0
		20	324/206	1.03	3.97	3.56	4.10	4.2	59	2.0
		22	484/308	0.69	9.44	5.63	6.53	4.5	61	2.1
		24	689/438	0.49	21.70	8.98	10.55	5.1	65	2.4
0.375 †	0.750	16	156/99	2.16	0.74	1.42	1.60	3.5	53	1.6
		18	224/143	1.50	1.71	2.26	2.57	3.9	56	1.8
		20	351/223	0.96	4.26	3.57	4.08	3.9	56	1.9
		22	528/336	0.64	10.19	5.66	6.49	4.1	58	2.0
		24	754/479	0.45	23.44	9.02	10.46	4.7	61	2.2
0.500 †	0.750	16	168/107	1.99	0.79	1.41	1.58	3.2	51	1.5
		18	238/152	1.41	1.81	2.25	2.54	3.6	53	1.7
		20	378/240	0.89	4.53	3.55	4.02	3.6	53	1.7
		22	572/363	0.59	10.89	5.62	6.37	3.7	54	1.8
		24	819/520	0.41	25.14	8.97	10.28	4.2	57	2.0

spacer (in)	core (in)	AWG	turns in/out	I (A)	R_{hot} (Ω)	V_{20} (V)	V_{hot} (V)	P_{hot} (W)	T_{coil} ($^{\circ}\text{C}$)	J (A/mm 2)
0.625	0.750	16	180/115	1.86	0.85	1.42	1.58	2.9	49	1.4
		18	259/165	1.29	1.94	2.24	2.51	3.2	51	1.6
		20	414/263	0.81	4.92	3.56	3.98	3.2	51	1.6
		22	616/391	0.54	11.63	5.62	6.32	3.4	52	1.7
		24	884/561	0.38	26.86	8.97	10.18	3.9	55	1.8
0.750	0.750	16	192/122	1.76	0.90	1.43	1.58	2.8	47	1.3
		18	280/178	1.21	2.08	2.26	2.52	3.0	49	1.5
		20	441/280	0.77	5.21	3.59	4.00	3.1	49	1.5
		22	660/419	0.51	12.39	5.68	6.35	3.3	50	1.6
		24	949/603	0.36	28.67	9.06	10.21	3.6	53	1.7
0.875	0.750	16	210/134	1.61	0.98	1.43	1.57	2.5	45	1.2
		18	301/192	1.12	2.23	2.27	2.51	2.8	47	1.4
		20	468/297	0.72	5.50	3.59	3.98	2.9	47	1.4
		22	704/447	0.48	13.13	5.68	6.31	3.0	48	1.5
		24	1014/644	0.33	30.40	9.07	10.15	3.4	51	1.6

B Reproducing this report

From the repository root:

```
cd gmsb_getdp/assembly
python emag_sweep.py --validate           # linear vs committed nonlinear, baseline
python emag_sweep.py                     # full grid, resumable (~1 h)
python emag_sweep.py --nonlinear h1.125_ci0.750_oo1.500
cd ../../
python report/render_3d.py
python report/make_report.py              # figures, tables, PDF (needs tools/tectonic)
```